



Generative Artificial Intelligence Chatbots in Mental Health: Current Evidence, Clinical Applications, Risks, and Future Perspectives

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Abstract

This narrative review aims to evaluate the use of generative artificial intelligence (AI) and large language model (LLM)-based chatbots in mental health care in light of current scientific evidence. The findings indicate that these systems may be effective in reducing symptoms of depression, anxiety, and stress, as well as in supporting psychoeducation and improving self-management skills. In addition, their 24/7 availability, anonymity, and low-cost nature suggest significant potential for improving access to mental health services.

However, the effectiveness of these tools varies depending on intervention type, user characteristics, and clinical context, with heterogeneous long-term outcomes reported across studies. Major limitations in clinical application include data privacy concerns, ethical responsibility, algorithmic bias, insufficient crisis management capabilities, and legal uncertainty. In particular, their use as standalone tools in high-risk psychiatric conditions is not considered appropriate.

In conclusion, generative AI-based chatbots should be regarded as supportive and complementary tools rather than replacements for human therapists. Future research should focus on long-term effectiveness, safety, and real-world clinical integration of these systems.

Keywords

Generative artificial intelligence; large language models; chatbots; mental health; depression; psychoeducation; cognitive behavioral therapy; ethics; data security.

1. Introduction

Mental health disorders are among the most important public health problems of the present day because of their increasing prevalence worldwide, their negative effects on quality of life, and the economic burden they impose on healthcare systems. Although depression, anxiety disorders, and stress-related mental health problems affect millions of people worldwide, many individuals are unable to access the psychological support they need in a timely manner because of factors such as the shortage of mental health professionals, long waiting times, high costs, and the stigma associated with mental illnesses. This situation has increased the need for accessible and scalable digital solutions that can support mental health services.

[2,3,18,19]

Recent advances in the field of generative artificial intelligence (GenAI) have made it possible to use chatbots developed based on Large Language Models (LLMs) in the field of mental health. In addition to general-purpose systems such as ChatGPT, Gemini, and Claude, mental health-focused chatbots such as Woebot, Wysa, and Tess are able to communicate with users in a context-sensitive manner, provide psychoeducation, support cognitive behavioral therapy-based interventions, and offer personalized feedback through their natural language processing capabilities. Owing to their continuous accessibility, their ability to provide anonymous use, and their potential to reduce concerns related to stigma, these systems are considered promising digital support tools, particularly for individuals with limited access to professional mental healthcare. [2,3,5,8,17,19]

Current studies show that generative artificial intelligence chatbots may contribute to reducing symptoms of mild to moderate depression, anxiety, and stress and may support therapeutic processes. However, hallucinations (the generation of inaccurate information), algorithmic biases, the inability to adequately assess crisis situations, data privacy and security concerns, and uncertainties regarding ethical and legal responsibilities emerge as the main challenges to the clinical use of these technologies. Therefore, the role of generative artificial intelligence chatbots in mental health should be comprehensively evaluated in light of the current scientific evidence. [1,2,3,6,7,8,9,10,12,15,19,20]

The aim of this narrative review is to summarize the current evidence regarding generative artificial intelligence chatbots in mental health, evaluate their clinical application areas and potential benefits, discuss the associated risks and ethical issues, and provide a holistic perspective for future research and clinical practice.

2. Methods

This study is a narrative review prepared to evaluate the current scientific evidence regarding the use of generative artificial intelligence (GenAI) and large language model (LLM)-based chatbots in the field of mental health. All figures and graphs presented in this study were originally prepared for the purpose of synthesizing and visualizing the scientific evidence based on the data obtained through the literature review.

The literature search was conducted in the PubMed/MEDLINE database by considering studies published between January 2018 and June 2026. During the search process, the keywords "generative artificial intelligence," "large language model," "chatbot," "mental health," "psychiatry," "psychology," "depression," and "anxiety" were searched in different combinations using Boolean operators (AND, OR).

Original studies, randomized controlled trials, systematic reviews, and meta-analyses examining the clinical use, effectiveness, safety, and ethical aspects of generative artificial intelligence- or large language model-based chatbots in the field of mental health were

included in this review. Studies that were not directly related to mental health or were not considered compatible with the aim of the review were excluded from the evaluation.

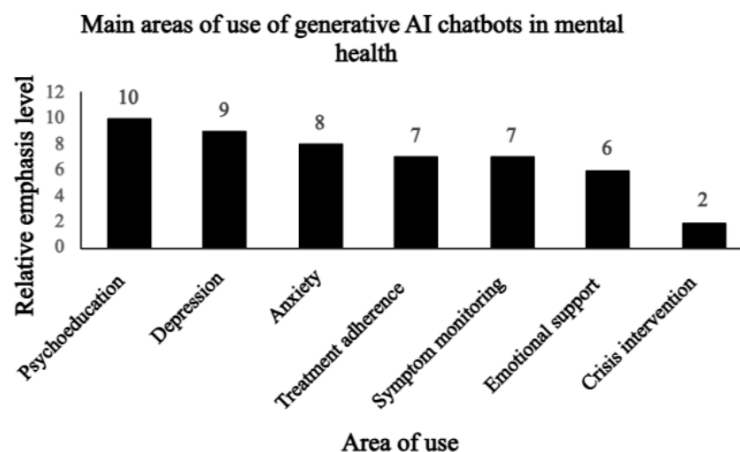
The findings obtained from the literature were evaluated using a narrative synthesis approach under the headings of clinical applications, effectiveness, psychoeducation, risks, and ethical and legal issues. This study is not a systematic review or meta-analysis but rather a narrative review aiming to critically and comprehensively summarize the current literature.

3. Literature Findings

3.1. Areas of Use of Generative Artificial Intelligence Chatbots in Mental Health

Generative artificial intelligence-based chatbots are used in mental health in various areas, including psychological support, psychoeducation, symptom monitoring, and supporting cognitive behavioral therapy-based interventions. In particular, systematic reviews conducted in young populations show that chatbot-based interventions can reduce stress levels and alleviate psychological distress [1]. However, it has been reported that the magnitude of these effects is generally small to moderate and varies depending on the intervention design [1,4].

Generative artificial intelligence and large language models are considered supportive tools for clinical processes in mental health services. Owing to their natural language understanding and generation capabilities, these systems are able to interact with users, provide psychoeducation, and offer basic emotional support. However, the literature emphasizes that these technologies still have important limitations, particularly with respect to clinical safety, accuracy, and crisis management, and that they remain at an early stage of development [2,5]. The relative distribution of the importance of these areas of use in the literature is summarized in Graph 1.



Graph 1. Major areas of use of generative artificial intelligence chatbots in mental health.

Note: The values presented in the figure do not represent statistical measurements but rather the relative levels of importance based on the qualitative synthesis of the literature reviewed in this narrative review.

Systematic reviews demonstrate that generative artificial intelligence models perform particularly well in psychoeducation, promoting emotional awareness, and supportive interventions. In contrast, their performance is reported to be limited in diagnosis, clinical decision support processes, and situations requiring cultural sensitivity. In addition, users have reported various concerns regarding the reliability, level of empathy, and accuracy of these systems [3,5].

Clinical studies and randomized controlled trials indicate that chatbot-based interventions can reduce symptoms of depression and anxiety. More pronounced effects have been reported particularly in individuals with early-stage and mild-to-moderate symptoms, and some studies have reported high levels of user satisfaction [1,4]. However, it has also been noted that clinical effectiveness is not consistent across all populations and that significant differences remain limited in some studies.

Overall, the literature shows that generative artificial intelligence chatbots are most commonly used in mental health for psychoeducation, emotional support, symptom monitoring, and therapy-supportive applications. However, considerable uncertainties remain regarding clinical safety, ethical responsibilities, data privacy, and long-term effectiveness. Therefore, it is emphasized that future studies should be conducted in real-world clinical settings using more robust methodological designs [2,3,5].

3.2. Effectiveness in the Management of Depression, Anxiety, and Stress

The effects of artificial intelligence-based chatbots and conversational agents on symptoms of depression and anxiety have generally been reported as significant and favorable in current systematic reviews and meta-analyses. These interventions have been shown to reduce symptoms of depression and anxiety, particularly during short-term use [6,7].

Findings from meta-analyses indicate small to moderate effect sizes for depression, while similar but more limited improvements have been observed for anxiety symptoms [6,7]. However, some studies have reported that these effects may diminish or may not be sustained during long-term follow-up [6].

When cognitive behavioral therapy (CBT)-based artificial intelligence chatbots are examined, more consistent and pronounced improvements in symptoms of depression and anxiety are observed. Applications such as Woebot, Wysa, and Youper have been reported to reduce both

depression and anxiety levels while providing high user satisfaction and therapeutic engagement [8].

Randomized controlled trials conducted in clinical settings demonstrate that artificial intelligence-assisted systems can achieve higher improvement rates in symptoms of depression and anxiety compared with standard treatment. In addition, these systems have been reported to increase treatment engagement and improve process efficiency by supporting clinical workflows [10].

Although the findings regarding stress management and overall psychological well-being are more heterogeneous, artificial intelligence-based interventions have been reported to reduce stress-related symptoms, particularly in short-term applications. However, their effectiveness with respect to anxiety and overall psychological well-being has been reported to be limited in some studies and to vary according to individual differences [11].

Studies conducted in young populations have shown that artificial intelligence-assisted conversational agents produce more pronounced effects on depression, whereas their effectiveness in the areas of anxiety, stress, and overall psychological well-being is more limited. In particular, greater improvements in depressive symptoms have been reported among subclinical populations [9].

Overall, the literature indicates that artificial intelligence-based chatbots are effective in reducing symptoms of depression and anxiety; however, their effects on stress management and overall psychological well-being are more variable. This suggests that their effectiveness may differ according to the type of intervention, duration, user characteristics, and clinical context [6,7,8,9,10,11].

3.3. Contributions to Psychoeducation, Self-Management, and Therapeutic Processes

Artificial intelligence-based chatbots and large language model (LLM)-based conversational agents have emerged as important digital tools that not only reduce symptoms of depression and anxiety but also support psychoeducation and self-management processes [6,12]. By providing structured content based on cognitive behavioral therapy (CBT), these systems help individuals understand the relationship between thoughts, emotions, and behaviors and enable the psychoeducational process to be maintained continuously [8,14].

Systematic reviews indicate that the regular use of artificial intelligence-assisted conversational agents enhances self-awareness, emotion monitoring, and cognitive restructuring skills [7,9]. Particularly in mobile- and messaging-based systems, the ability of users to monitor their daily mood and receive feedback is considered one of the key mechanisms that strengthen self-management capacity [7,8]. This process facilitates the

transition of users from passive recipients of care to active participants, thereby increasing their integration into the therapeutic process.

Another important dimension of the psychoeducational contributions of artificial intelligence-based interventions is their ability to improve treatment adherence and engagement. Clinical platforms and chatbot-based interventions contribute to users completing more sessions and maintaining participation in therapy for longer periods. For example, a randomized controlled trial using an artificial intelligence-assisted clinical platform reported higher attendance rates at therapy sessions and more pronounced reductions in symptoms of depression and anxiety [10]. This finding suggests that artificial intelligence provides not only direct psychological effects but also an indirect mechanism that enhances treatment adherence.

The new generation of LLM-based artificial intelligence agents has also been shown to provide more advanced cognitive interaction capabilities for psychoeducation and therapeutic processes. A 28-day randomized controlled trial reported that LLM-based dialogue agents significantly reduced symptoms of depression and anxiety and that their effectiveness increased particularly with repeated interactions. This finding indicates that sustained interaction plays a critical role in the development of self-management skills [12].

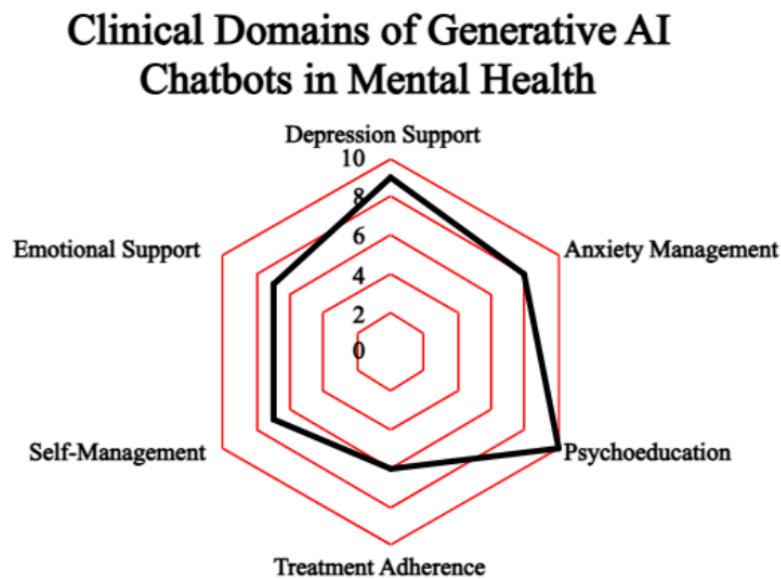
Similarly, LLM-based architectures used in psychotherapeutic interactions have been shown to enhance the delivery of clinically structured cognitive behavioral therapy and improve the quality of therapeutic communication. These systems have been reported to generate therapeutic responses that approach clinician performance and, according to some measures, even surpass it [13]. This suggests that artificial intelligence may assume not only a supportive role but also a role in structuring the therapeutic process.

Studies conducted in young adult populations also support these findings. Cognitive behavioral therapy (CBT)-based artificial intelligence chatbots have been reported to be effective in reducing depression and loneliness, with these effects being particularly pronounced among individuals with higher levels of stress. In addition, even short-term interventions have been shown to be effective in reducing cognitive distortions and strengthening coping skills through psychoeducational content [11].

Web- and mobile-based chatbot interventions also contribute to self-management processes. In particular, therapy chatbots used in subclinical populations have been reported not only to reduce symptoms of depression and anxiety but also to increase the frequency with which users apply self-help strategies. Furthermore, it has been emphasized that a higher frequency of use enhances therapeutic effects and is directly associated with the effectiveness of self-management skills [14].

In conclusion, artificial intelligence-based chatbots and LLM-assisted conversational agents demonstrate multidimensional contributions by providing psychoeducation, enhancing self-

management skills, increasing engagement in therapy, and supporting the structuring of clinical processes [6,7,8,9,10,11,12,13,14]. The principal clinical applications of these contributions in mental health services are schematically summarized in Graph 2.



Graph 2. Clinical applications of generative artificial intelligence chatbots in mental health. *Note: The values presented in the figure do not represent statistical measurements but rather the relative levels of importance based on the qualitative synthesis of the literature reviewed in this narrative review.*

These systems are positioned not as replacements for traditional psychotherapy but as digital health tools that complement it by improving accessibility and promoting greater continuity of the therapeutic process.

3.4. Clinical Applications and Use in Therapeutic Processes

Artificial intelligence (AI)-based systems are increasingly being used in mental health services not only as interventions that reduce symptoms but also as supportive tools that transform the delivery of clinical practice. One of the most important contributions of these systems is their ability to support clinical workflows during psychotherapy, thereby reducing therapists' workload and making the treatment process more systematic. In particular, AI-based platforms can automatically transcribe therapy sessions, generate summaries, and convert them into clinical notes, substantially reducing therapists' administrative burden. This enables clinical processes to be conducted more efficiently while allowing therapists to devote more time to patient care. Indeed, a randomized controlled trial evaluating an AI-assisted therapy platform demonstrated that therapists completed progress notes in significantly less time than with standard care, resulting in a marked improvement in clinical workflow efficiency [10].

In addition, artificial intelligence systems enhance the quality of psychotherapeutic processes by providing therapists with feedback through clinical decision support mechanisms. By analyzing session content, these systems can evaluate the use of evidence-based therapeutic techniques and provide clinicians with objective data regarding the therapeutic process. Consequently, psychotherapy can be informed not only by subjective clinical observations but also by data-driven evaluation. Similarly, AI-based conversational agents have been shown to improve clinical competencies by analyzing therapeutic interactions and to contribute to the delivery of high-quality cognitive behavioral therapy [13]. These findings suggest that artificial intelligence may function not only as a supportive tool but also as a mechanism for monitoring and improving the quality of the therapeutic process.

Another important contribution of AI in clinical practice is its potential to facilitate depression management through integration into primary care services. Chatbots provide patients with accessible, 24/7 support, enabling early intervention and reducing barriers to accessing mental health care. The integration of these systems into primary care not only facilitates patient follow-up but also helps bridge the gap in settings where access to mental health specialists is limited [17]. Furthermore, AI-based chatbots have been highlighted as promising tools for integration into healthcare systems because of their capabilities in early symptom detection, personalized feedback, and continuous support for depression management [17,18].

Nevertheless, the integration of AI into clinical practice also presents several important challenges. In particular, the complex and subjective nature of mental disorders highlights the limitations of artificial intelligence systems with respect to clinical accuracy and reliability. Some studies emphasize that current AI systems are not sufficiently capable of making clinical diagnoses or replacing human therapists and should therefore be regarded only as supportive tools [16]. In addition, ethical concerns, data privacy, algorithmic bias, and the need to preserve the human dimension of the therapeutic relationship remain major issues that must be carefully addressed in clinical practice [15,18].

Overall, the literature indicates that AI-based systems provide multifaceted support in clinical psychotherapy by both reducing clinicians' workload and improving the quality of treatment. However, the integration of these technologies into clinical practice should be viewed as a complementary approach that supports clinicians' decision-making processes and makes

psychotherapy more accessible and efficient, rather than replacing human clinicians [15,18]. The potential clinical integration pathway of generative artificial intelligence-based chatbots in mental health services is conceptually illustrated in Figure 1.

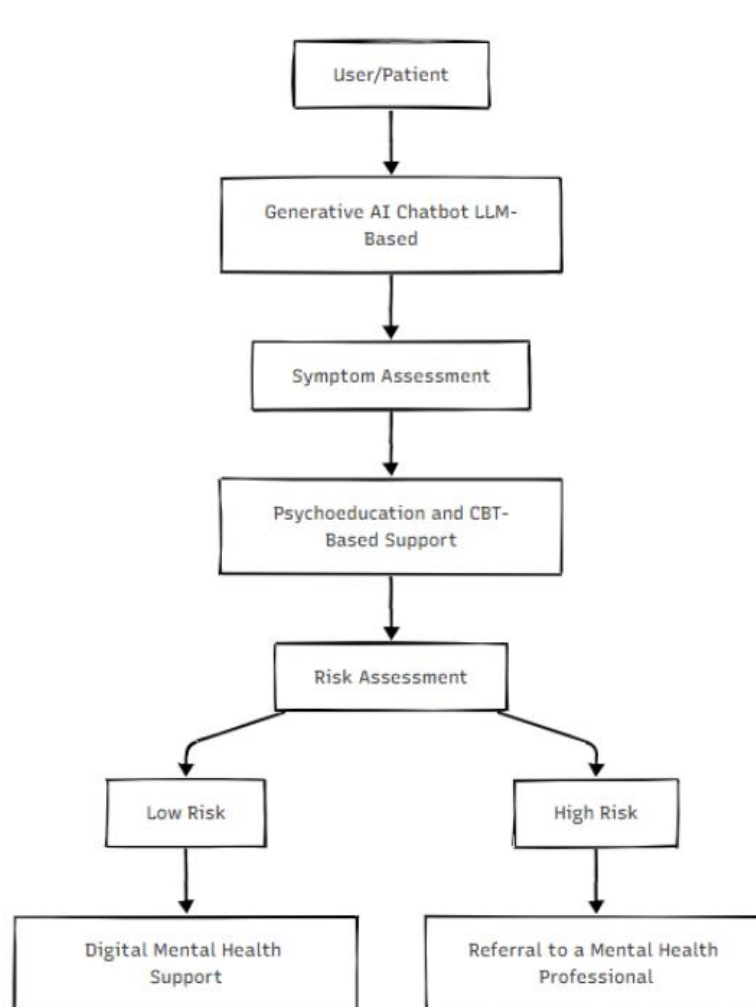


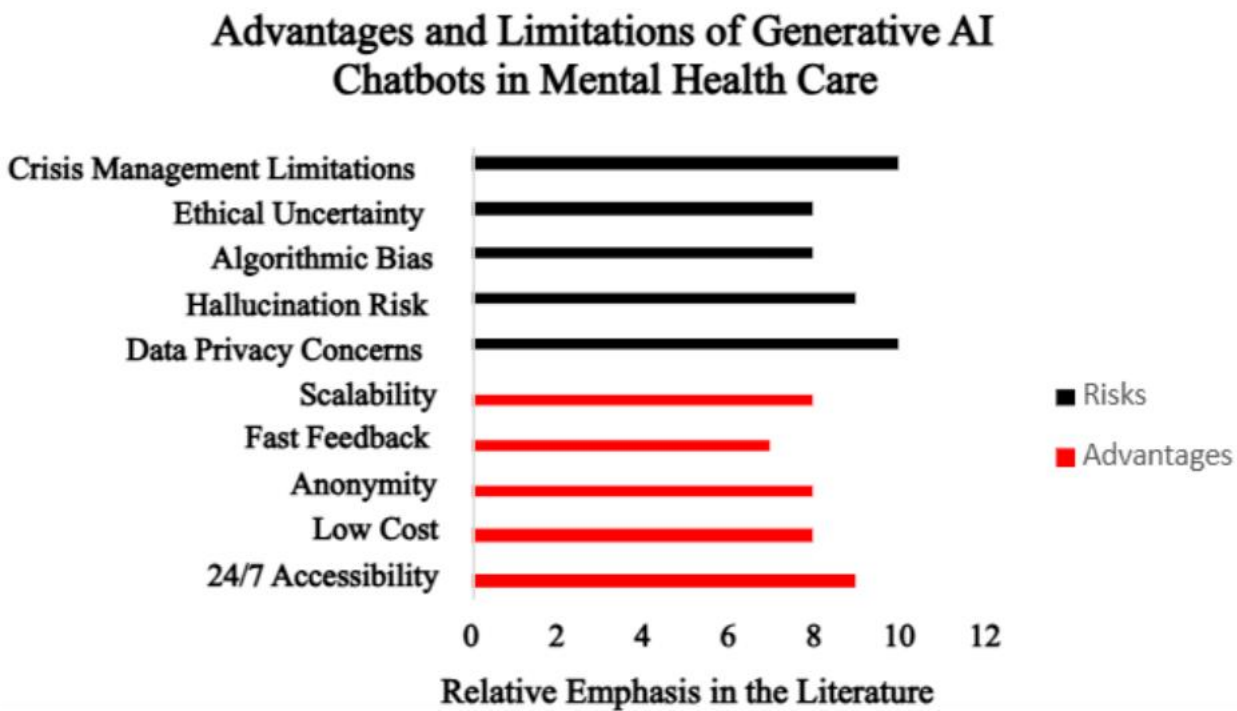
Figure 1. Conceptual clinical workflow of generative artificial intelligence-based chatbots in mental health services.

3.5. Findings Related to Ethics, Legal Issues, and Data Security

The widespread adoption of artificial intelligence (AI), particularly generative artificial intelligence (GenAI)-based systems, in mental health has raised important concerns not only regarding clinical effectiveness but also with respect to ethical, legal, and data security issues. Although the literature recognizes the potential of these technologies to improve access to mental health services, reduce clinical workload, and facilitate psychological support, it also

highlights significant structural challenges related to ensuring that these systems are implemented in a safe, equitable, and accountable manner [18,19].

One of the most critical concerns is data privacy and security. Because AI-based chatbot systems continuously interact with users, they collect large volumes of sensitive personal data. This raises concerns about data breaches, unauthorized use, and the sharing of information with third parties, particularly in highly sensitive clinical domains such as mental health [19,20]. The principal advantages and limitations of generative artificial intelligence chatbots identified in the literature are comparatively presented in Graph 3. Current studies indicate that data protection mechanisms remain insufficiently transparent in many systems and that users are provided with limited information regarding how their data are processed. This issue represents a significant risk, particularly in regions where legal frameworks comparable to the General Data Protection Regulation (GDPR) are absent or inadequately enforced.



Graph 3. Principal advantages and limitations of generative artificial intelligence chatbots in mental health. *Note: The values presented in the figure do not represent statistical measurements but rather the relative levels of importance based on the qualitative synthesis of the literature reviewed in this narrative review.*

A second major issue concerns ethical responsibility and clinical accountability. The literature highlights the risk that artificial intelligence systems may generate inaccurate, incomplete, or contextually inappropriate responses, commonly referred to as *hallucinations* [18,20]. In high-risk fields such as mental health, this increases the potential for patient harm. One of the central debates concerns responsibility when such errors occur: whether accountability should rest with the system developer, the healthcare provider, or the clinician

using the system. Current studies emphasize that no clear legal framework currently exists to address this issue and that accountability mechanisms remain insufficiently defined [20].

A third major ethical concern involves safety, the risk of harm, and crisis management. In particular, AI systems may be unable to provide adequate clinical intervention in cases involving suicide risk, self-harm, or acute psychiatric crises [17,20]. Although some chatbot-based systems have demonstrated supportive benefits, the literature indicates that the risk of providing inappropriate guidance or inadequate responses during crisis situations may have serious clinical consequences [14,17]. Consequently, AI systems should not be used as independent therapeutic decision-makers but rather should be regarded as supportive tools within clinical practice.

A fourth major concern relates to bias, fairness, and equity. The use of socio-culturally unbalanced training datasets in artificial intelligence models may result in inaccurate or inadequate outputs for certain population groups [18,19]. This may exacerbate digital inequalities in access to mental health services and further widen existing health disparities. The literature also indicates that individuals with lower levels of digital literacy tend to benefit less from these systems, giving rise to concerns regarding digital ethical equity [20].

A fifth important ethical dimension concerns transparency, explainability, and user trust. The inability of users and clinicians to fully understand how "black-box" AI models generate their decisions may undermine confidence in these systems [20]. This issue is particularly relevant in psychotherapy, where trust is fundamental to the therapeutic alliance and may be compromised by insufficient transparency.

Finally, the literature strongly emphasizes the lack of comprehensive legal regulation and the need for robust ethical governance. Current evidence indicates that no comprehensive and standardized international legal framework has yet been established for AI systems used in mental health care [18,19]. Accordingly, clear standards are needed for data security, patient rights, algorithmic accountability, and clinical oversight. Furthermore, ethical frameworks should focus not only on technical safety but also on preserving the principles of human-centered care.

Overall, the literature indicates that AI-supported mental health applications offer significant opportunities while also presenting multidimensional risks related to data privacy, ethical responsibility, clinical safety, fairness, and legal uncertainty [18,19,20]. Therefore, the integration of these technologies into clinical practice is considered feasible only when supported by robust ethical guidelines, appropriate regulatory frameworks, and continuous human oversight.

4. Discussion

This narrative review suggests that generative artificial intelligence (GenAI) and large language model (LLM)-based chatbots may play a supportive role in mental health services, particularly in the management of mild to moderate symptoms of depression and anxiety. However, the current literature indicates that the effectiveness of these technologies varies according to factors such as intervention duration, the AI model used, the target population,

and the clinical context. Therefore, it is not realistic to assume that generative AI systems will provide the same level of benefit across all user groups [1,2,6,7].

A consistent finding across the studies included in this review is that AI-supported chatbots have demonstrated favorable outcomes, particularly in psychoeducation, symptom monitoring, and cognitive behavioral therapy (CBT)-based interventions. Nevertheless, the limited evidence regarding long-term effectiveness, together with the methodological heterogeneity across studies, requires that these positive findings be interpreted with caution. Differences in sample characteristics, intervention duration, and the chatbot systems employed may account for much of the variability observed among study findings [6,7,8,9].

Another important finding is the potential of generative AI systems to improve access to mental health services. Their continuous availability, relatively low cost, and ability to reduce concerns related to stigma offer substantial advantages, particularly for individuals who face barriers to obtaining professional psychological support. However, increased accessibility should not be equated with clinical safety. At present, there is insufficient evidence to support replacing human clinical judgment in situations involving suicide risk, acute psychiatric conditions, or other complex mental health presentations [2,5,17,18].

Another prominent issue identified in the literature concerns ethical and legal challenges. Uncertainties related to data privacy, algorithmic bias, the generation of inaccurate information (*hallucinations*), and accountability remain among the principal barriers to the safe integration of generative artificial intelligence systems into clinical practice. Accordingly, future efforts should focus not only on technological advancements but also on strengthening ethical standards, regulatory frameworks, and mechanisms for clinical oversight [5,15,18,19,20].

Overall, the current body of scientific evidence suggests that generative artificial intelligence-based chatbots are more appropriately viewed as digital tools that support, rather than replace, mental health professionals. Future research should prioritize larger study populations, long-term follow-up, and evaluations conducted in real-world clinical settings to strengthen the evidence base regarding the safety and effectiveness of these technologies. Based on the findings of this narrative review, a conceptual framework summarizing these relationships is presented in Figure 2.

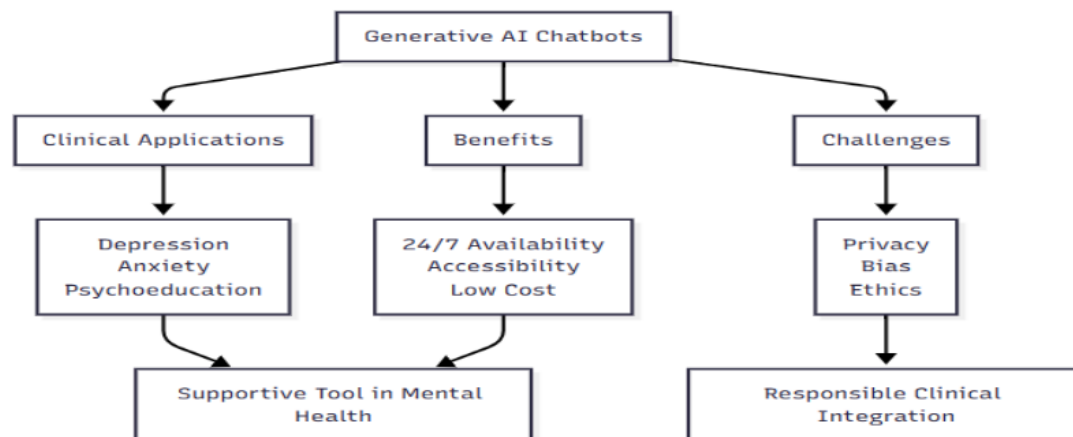


Figure 2. Conceptual framework of generative artificial intelligence chatbots in mental health services.

5. Conclusion

This narrative review comprehensively evaluated the growing use of generative artificial intelligence (GenAI) and large language model (LLM)-based chatbots in mental health care. The findings indicate that these systems have considerable potential to reduce symptoms of depression, anxiety, and stress, strengthen psychoeducational interventions, and support self-management skills. Furthermore, their continuous availability, relatively low cost, anonymity, and potential to reduce stigma suggest that they may help alleviate some of the structural barriers to accessing mental health services.

However, the current evidence demonstrates that the effectiveness of these technologies varies according to the clinical context, user characteristics, and intervention design. While more pronounced benefits have been reported for short-term interventions and individuals with mild to moderate symptoms, the evidence regarding long-term effectiveness and improvements in overall psychological well-being remains limited and heterogeneous. In addition, LLM-based systems continue to face important limitations in areas such as clinical accuracy, crisis management, and cultural sensitivity.

Among the most critical limitations are concerns regarding data privacy and security, uncertainty surrounding ethical responsibility, the risk of algorithmic bias, and the lack of comprehensive legal regulation. In particular, in high-risk clinical situations such as suicide risk and acute psychiatric emergencies, these systems cannot be considered safe as standalone interventions and should always be used under the supervision of qualified mental health professionals. These findings indicate that generative artificial intelligence should be positioned not as an independent therapist but as a supportive and complementary tool within mental health care.

Consequently, although generative artificial intelligence–based chatbots offer significant opportunities in mental health care, their safe and effective clinical implementation currently requires robust ethical frameworks, regulatory mechanisms, and comprehensive clinical validation studies. Future research should focus not only on effectiveness outcomes but also on long-term safety, real-world applications, and human–artificial intelligence collaboration models to facilitate the sustainable and safe integration of these technologies into mental health care.

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Conflict of Interest Statement

No conflict of interest has been declared.